

Example 6

Solve the linear system

$$\begin{aligned}x + 2y + z &= 4 \\3x + 8y + 7z &= 20 \\2x + 7y + 9z &= 23.\end{aligned}\tag{14}$$

Solution First, we add -3 times the first equation to the second equation; the result is

$$\begin{aligned}x + 2y + z &= 4 \\2y + 4z &= 8 \\2x + 7y + 9z &= 23.\end{aligned}$$

Then addition of -2 times the first equation to the third equation yields

$$\begin{aligned}x + 2y + z &= 4 \\2y + 4z &= 8 \\3y + 7z &= 15.\end{aligned}$$

We have now eliminated x from the second and third equations. To simplify the process of eliminating y from the third equation, we multiply the second equation by $\frac{1}{2}$, to obtain

$$\begin{aligned}x + 2y + z &= 4 \\y + 2z &= 4 \\3y + 7z &= 15.\end{aligned}$$

Finally, addition of -3 times the second equation to the third equation gives

$$\begin{aligned}x + 2y + z &= 4 \\y + 2z &= 4 \\z &= 3.\end{aligned}\tag{15}$$

This system has a *triangular* form that makes its solution easy. By back substitution of $z = 3$ in the second equation in (15), we find that

$$y = 4 - 2z = 4 - 2(3) = -2;$$

then the first equation yields

$$\begin{aligned}x &= 4 - 2y - z \\&= 4 - 2(-2) - (3) = 5.\end{aligned}$$

Thus our original system in (14) has the unique solution $x = 5$, $y = -2$, $z = 3$. 

Comment The steps by which we transformed (14) into (15) show that every solution of the system in (14) is a solution of the system in (15). But these steps can be reversed to show similarly that every solution of the system in (15) is also a solution of the system in (14). Thus the two systems are equivalent in the sense that they have the same solution set. The computation at the end of Example 6 shows that (15) has the unique solution $(x, y, z) = (5, -2, 3)$, and it follows that this also is the unique solution of the original system in (14). 

Example 7

Solve the system

$$\begin{aligned}3x - 8y + 10z &= 22 \\x - 3y + 2z &= 5 \\2x - 9y - 8z &= -11.\end{aligned}\tag{16}$$

Solution In order to avoid fractions in the elimination of x , we first interchange the first two equations to get

$$\begin{aligned}x - 3y + 2z &= 5 \\3x - 8y + 10z &= 22 \\2x - 9y - 8z &= -11.\end{aligned}$$

Addition of -3 times the first equation to the second gives

$$\begin{aligned}x - 3y + 2z &= 5 \\y + 4z &= 7 \\2x - 9y - 8z &= -11,\end{aligned}$$

and then addition of -2 times the first equation to the third equation gives

$$\begin{aligned}x - 3y + 2z &= 5 \\y + 4z &= 7 \\-3y - 12z &= -21.\end{aligned}$$

Finally, addition of 3 times the second equation to the third equation yields

$$\begin{aligned}x - 3y + 2z &= 5 \\y + 4z &= 7 \\0 &= 0.\end{aligned}\tag{17}$$

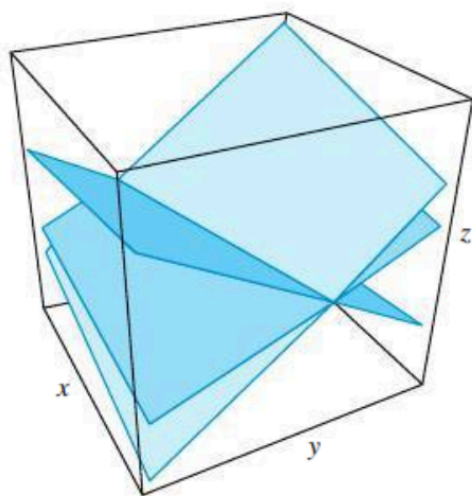


FIGURE 3.1.4. The three planes of Example 7.

Because the third equation has disappeared, we can choose $z = t$ arbitrarily and then solve for x and y :

$$\begin{aligned}y &= 7 - 4z = 7 - 4t, \\x &= 5 + 3y - 2z \\&= 5 + 3(7 - 4t) - 2t = 26 - 14t.\end{aligned}$$

Thus our original system in (16) has infinitely many solutions. Moreover, one convenient way of describing them is this:

$$x = 26 - 14t, \quad y = 7 - 4t, \quad z = t. \quad (18)$$

The arbitrary parameter t can take on all real number values, and in so doing generates all the (infinitely many) solutions of the original system in (16). Figure 3.1.4 shows the three planes corresponding to the three equations in (16). These planes intersect in a straight line that is parameterized by the equations in (18). ■